1 Data

ABIDE I (PCP)
1,112 subjects
CPAC pipeline,
band-pass, no GSR

→

Quality check
time series present,
no rater marked fail
→ 873 subjects

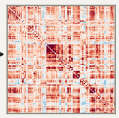
→

Length filter
≥ 100 time points
→ 848 subjects:
393 autism, 455 controls,
19 sites

→

Connectome
AAL atlas, 116 regions
Pearson r per region pair
Fisher z, zero diagonal

→



x_i: 116 × 116
brain graph:
nodes = regions,
edges = 6,670
region pairs

2 Benchmark construction

Input: the 848 connectomes (455 controls, 393 autism)

Disease effect: the known answer

1 Choose disease regions
ranking set: site-stratified 20% of controls + all autism subjects
Cohen's d per edge → top 10 regions by mean |d| → truth: 45 edges

2 Fake labels
planting pool: the other 343 controls with no empty region
in each site, a random half labelled 'patient' (y = 1), per split

3 Plant the effect (fake patients only)
shared random signal, orthogonal to the subject's own signals,
added to the 10 regions' signals; connectivity recomputed

4 Calibrate the strength ℓ
mean shift of the 45 edges = ℓ × median real autism effect
$\ell \in \{1, 2, 4\}$; edges to other regions: 'don't care'

Scanner effect: the known trap

1 Choose scanner regions
η^2 between sites per edge (controls; sites with ≥ 20 controls)
top 10 regions outside the disease set → 45 scanner edges

2 Size from real hospitals
median over site pairs of the mean |difference of site means|
on the scanner edges: a realistic hospital difference

3 Scanner-B version of every subject
main: shared signal added to the 10 regions (person-specific)
robustness: one fixed shift per edge (ComBat's own model)

4 Two scans per subject
x_i at scanner A and x_i^B at scanner B, same label y_i
the true pair is what oracle CF-Site trains with

Link the scanner to the label, per fold
training and validation sites: scanner B for a fraction ρ of fake patients and $1-\rho$ of fake controls (ρ = 0.5 to 0.9); the test site is always 50/50,
so at test the scanner carries no information about the label → benchmark data (x_i, y_i, b_i)

3 Protocol, fixes, training

Evaluation protocol
BrainOOD: 10 leave-site-out folds
each fold: 1 test site, 1 validation site, training on the rest
recovery, drift: 3 splits × 3 seeds
fixes: 6 splits × 3 seeds
every choice on validation data;
test sites never used for a choice

Fixes compared (one per run, fitted on the training subjects of each fold)

Plain
train on x as it is

ComBat (4 versions)
scanner A/B as batch
EB shrinkage: yes or no
diagnosis kept: yes or no
test: covariate at mean

CF-Site (ours)
$\tilde{x} = x$ at the other scanner
(label-protected ComBat; EB-swap variant)
loss: CE on x and on $\tilde{x}$

Oracle CF-Site
$\tilde{x}$ = the true x^B scan
not available in practice:
an upper bound

Training (neural models): Adam, learning rate 10^{-3}, weight decay 10^{-3}, batch 32, 150 epochs;
weights of the epoch with the lowest validation loss after a 30-epoch warm-up;
inputs standardised inside each model with the training mean and SD of every entry

4 Model architectures

Models f: the raw 116 × 116 matrix in, two class scores out (patient vs. control)

Logistic regression
upper triangle: 6,670 edges
standardise each edge
L2 logistic regression
C ∈ {0.01, 0.1, 1} on validation
explanation: |coefficients|

MLP
upper triangle: 6,670 edges
dropout 0.5
linear 6,670 → 64, ReLU
dropout 0.5
linear 64 → 2

GCN (BrainGB set-up)
node features: rows of x
graph: top 20% |r|, w = ReLU(r)
self-loops, $D^{-1/2}WD^{-1/2}$
GraphConv 116 → 16 + root
GraphConv 16 → 16 + root
concat 116 × 16 → linear 2

BrainNetCNN
edge-to-edge 1 → 32
edge-to-edge 32 → 32
edge-to-node 32 → 64
node-to-graph 64 → 256
FC 256 → 128 → 30 → 2
leaky ReLU 0.33, dropout 0.5

Sanity test
×4 effect, no shortcut:
saliency AUPRC ≥ 0.5

pass: MLP, GCN, BrainNetCNN
(LR: 0.98 at ×2)
fail: BrainGNN, BNT
(reported, not used further)

5 Explain and score

Explain (test-site subjects)

Saliency (main)
$|\partial(z_1 - z_0)/\partial x|$ per entry

Integrated Gradients
32 steps, zero baseline

Input mask
150 steps; size, entropy terms

Edge scores
symmetrise: 6,670 per subject

Group explanation S
mean over the test subjects

Controls: shuffled training labels give each explainer's chance level;
a univariate t-test on the training subjects gives a model-free reference

Score and test

Disease AUPRC
S ranks the 45 true edges above untouched ones;
chance 0.008

Scanner share
share of the top 45 edges of S that are scanner edges

Accuracy
pooled over all test subjects

Aggregate
folds and seeds → one value per split

Statistics
95% bootstrap CI over splits; sign test;
paired t-test; Holm

Decision
gain: CI excludes 0 and Holm $p < 0.05$;
otherwise a tie

6 Experiments

Recovery (Sec. 5.1)
ℓ = 1, 2, 4; ρ = 0.5
all explainers;
shuffled labels; LR

Drift (Sec. 5.2)
ℓ = 2; ρ = 0.5 → 0.9
MLP, GCN, LR,
BrainNetCNN

Fixes (Sec. 5.4)
ℓ = 2; ρ = 0.9
plain, 4 ComBat,
3 CF-Site, oracle

Robustness (Sec. 5.4)
fixed-shift scanner;
second region set;
IG; no small sites

Real data (5.3, 5.5)
no planting: hospital vs. autism (LR; AAL, CC200);
scanner map from training;
real labels; label probe